HK IMO Training 2021-2022
Problem Set 10
For 4 December 2021

Problem 37. Let P be a point inside $\triangle ABC$. Let D and E be the projections of P onto AB and AC respectively. Let F be the point on the line BC such that $DF = EF$. Let Q be the point such that F is the midpoint of PQ. Prove that $\angle BAP = \angle QAC$.

Problem 38. Let A and B be two sets of positive real numbers such that $|A| = |B| = 3$. It is known that A and B have the same sum of elements and the same product of elements. Also, $\min A \leqslant \min B$ and $\max A \geqslant \max B$ (where $\min A$ means the smallest element in A, etc.). Prove that $A = B$.

Problem 39. Let S be the set of all lattice points (x, y) in the coordinate plane with $1 \leqslant x, y \leqslant 2021$. At most how many points can be chosen from S so that there do not exist 4 chosen points forming a non-degenerate parallelogram (in any orientation)?

Problem 40. Define the sequence $\{a_n\}$ by $a_1 = 3$ and $a_{n+1} = a_n^3 + 3a_n$ for any positive integer n. Let p be an odd prime dividing a_n but not a_{n-1}, where $n > 1$. Prove that 3^{n-1} divides $p - 1$.